

Answer key

ASCII reference

Each character has a number (its *ASCII code*). Use this table wherever a question asks you to decode a message.

0	NUL	16	DLE	32	SP	48	0	64	@	80	P	96	`	112	p
1	SOH	17	DC1	33	!	49	1	65	A	81	Q	97	a	113	q
2	STX	18	DC2	34	"	50	2	66	B	82	R	98	b	114	r
3	ETX	19	DC3	35	#	51	3	67	C	83	S	99	c	115	s
4	EOT	20	DC4	36	\$	52	4	68	D	84	T	100	d	116	t
5	ENQ	21	NAK	37	%	53	5	69	E	85	U	101	e	117	u
6	ACK	22	SYN	38	&	54	6	70	F	86	V	102	f	118	v
7	BEL	23	ETB	39	'	55	7	71	G	87	W	103	g	119	w
8	BS	24	CAN	40	(	56	8	72	H	88	X	104	h	120	x
9	HT	25	EM	41	)	57	9	73	I	89	Y	105	i	121	y
10	LF	26	SUB	42	*	58	:	74	J	90	Z	106	j	122	z
11	VT	27	ESC	43	+	59	;	75	K	91	[	107	k	123	{
12	FF	28	FS	44	,	60	<	76	L	92	\	108	l	124	
13	CR	29	GS	45	-	61	=	77	M	93	]	109	m	125	}
14	SO	30	RS	46	.	62	>	78	N	94	^	110	n	126	~
15	SI	31	US	47	/	63	?	79	O	95	_	111	o	127	DEL

Problem 1.

Convert each binary number to decimal:

$$1011_2 = 8 + 2 + 1 = \mathbf{11}$$

$$1100_2 = 8 + 4 = \mathbf{12}$$

$$0111_2 = 4 + 2 + 1 = \mathbf{7}$$

$$1010_2 = 8 + 2 = \mathbf{10}$$

Hint: the four place values are 8, 4, 2, 1. Add up the place values wherever there is a 1.

Problem 2.

Convert each binary number to decimal:

$$10010_2 = 16 + 2 = \mathbf{18}$$

$$100000_2 = 32 = \mathbf{32}$$

$$111111_2 = 32 + 16 + 8 + 4 + 2 + 1 = \mathbf{63}$$

$$101101_2 = 32 + 8 + 4 + 1 = \mathbf{45}$$

Problem 3.

Convert each binary number to decimal:

$$01000001_2 = 64 + 1 = \mathbf{65}$$

$$00101010_2 = 32 + 8 + 2 = \mathbf{42}$$

$$10000000_2 = 128 = \mathbf{128}$$

$$11111111_2 = 128 + 64 + 32 + 16 + 8 + 4 + 2 + 1 = \mathbf{255}$$

With 8 bits the place values are 128, 64, 32, 16, 8, 4, 2, 1.

Problem 4.

Convert each decimal number to binary:

$$6 = 4 + 2 = 110_2$$

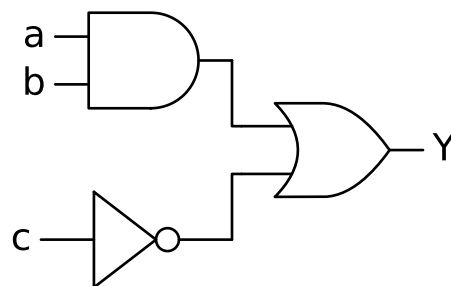
$$19 = 16 + 2 + 1 = 10011_2$$

$$42 = 32 + 8 + 2 = 101010_2$$

$$100 = 64 + 32 + 4 = 1100100_2$$

Hint: find the largest place value (... , 32, 16, 8, 4, 2, 1) that fits, subtract it, and repeat with what's left.

Problem 5.



Expression: $A \cdot B + \overline{C}$

Simplifies to: $\overline{C} + A \cdot B$

Truth table:

A	B	C	Y
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1

Problem 6.

The digit characters 0–9 are NOT the same as the numbers 0–9 — each has its own ASCII code.

Each character of a secret message has been written as its ASCII code in 8-bit binary. Convert each code to decimal, then use the ASCII table on the first page to decode the message.

binary	01000011	01010011	00110001	00110111
decimal	67	83	49	55
letter	C	S	1	7

Message: CS17

Problem 7.

Decode the secret word. Watch the capitalisation — uppercase and lowercase letters have different codes.

Each character of a secret message has been written as its ASCII code in 8-bit binary. Convert each code to decimal, then use the ASCII table on the first page to decode the message.

binary	01100010	01101001	01101110	01100001	01110010	01111001
decimal	98	105	110	97	114	121
letter	b	i	n	a	r	y

Message: binary

Problem 8.

Draw the truth table for:

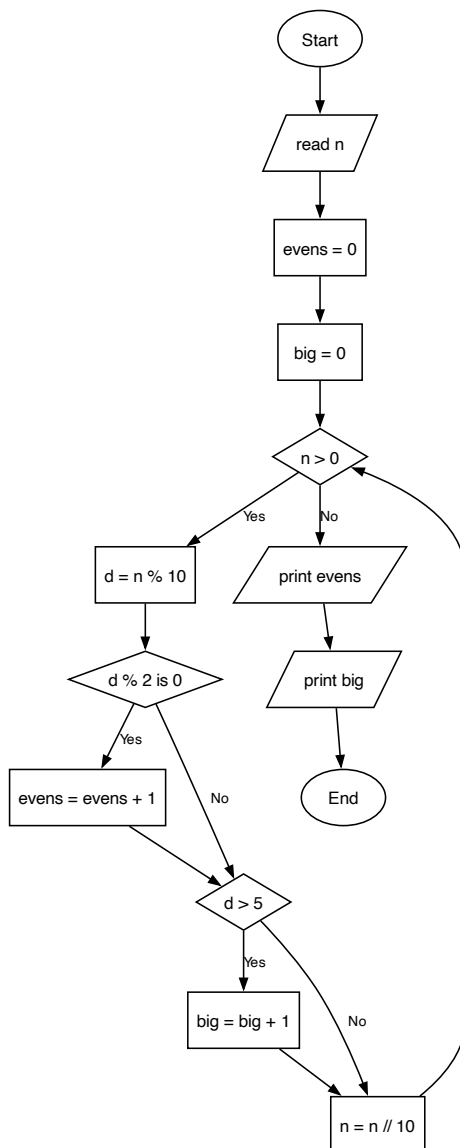
$$Y = A \oplus B \cdot C$$

A	B	C	Y
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

Problem 9.

Trace this algorithm for $n = 6391$, and complete the trace table.

Reminder: $n \% 10$ is the last digit of n , and $n // 10$ removes it (e.g. $6391 \% 10 = 1$ and $6391 // 10 = 639$).



n	evens	big	d	output
6391				
6391	0			
6391	0	0		
6391	0	0	1	
639	0	0	1	
639	0	0	9	
639	0	1	9	
63	0	1	9	
63	0	1	3	
6	0	1	3	
6	0	1	6	
6	1	1	6	
6	1	2	6	
0	1	2	6	
0	1	2	6	1
0	1	2	6	2

Output: 1, 2